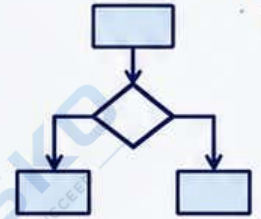


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Most Important Questions

INSTRUCTIONS



1. Understand the Concepts

Go through the basic concepts related to each question before practising.



2. Practice Regularly

Solve these important questions and their variations for better understanding.



3. Focus on Exam Pattern

These questions are selected as per previous year papers and latest syllabus.



4. Attempt Within Time Limit

Practice under timed conditions to improve speed and accuracy.



5. Revise and Self-Check

Review your answers, identify weak areas and revise regularly to score higher.

Learn • Practice • Improve • Succeed

YOUR HARD WORK PAYS OFF!



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C++ & Python — Most Important Questions

Unit I - OOP Concepts & Paradigm

- **Answers in one or two lines (1 mark questions)**

1. What is OOP?
2. What is the OOP paradigm?
3. State any two features of OOP.
4. What is data abstraction?
5. What is encapsulation?
6. What is a class?
7. What is an object?
8. Name any two OOP languages.
9. What is structured programming?

- **Short Answer Questions (3 marks questions)**

1. Explain the basic concepts of OOP.
2. Explain features, advantages and applications of OOP.
3. Differentiate between Structured Programming and OOP.
4. Explain data abstraction and encapsulation.
5. Explain classes and objects.

- **Long Answer Questions (6 marks questions)**

1. Explain OOP paradigm, its features, advantages and applications.
2. Differentiate between Structured Programming and Object-Oriented Programming.
3. Explain data abstraction, encapsulation, classes and objects with examples.
4. Explain trending OOP languages.

Unit II -C++ Functions, Constructors & Inheritance

- **Answers in one or two lines (1 mark questions)**

1. What is a function prototype?
2. What is an inline function?
3. What is a default argument?
4. What is function overloading?
5. What is a constructor?
6. What is a default constructor?
7. What is a parameterized constructor?
8. What is a copy constructor?
9. What is inheritance?
10. Name the types of inheritance.

- **Short Answer Questions (3 marks questions)**

1. Explain function prototype and inline function.
2. Explain default arguments and function overloading.
3. Explain types of constructors.
4. Explain default, parameterized and copy constructors.
5. Explain inheritance and its types.
6. Explain visibility modes in inheritance.

- **Long Answer Questions (6 marks questions)**

1. Explain function overloading with suitable examples.
2. Explain Default, Parameterized and Copy Constructors with examples.
3. Explain inheritance and its types: Single, Multiple, Hierarchical, Multilevel and Hybrid.
4. Explain visibility modes in inheritance.

Unit III - Introduction to Python

- **Answers in one or two lines (1 mark questions)**

1. What is Python?
2. Who developed Python?
3. State any two features of Python.
4. What is Anaconda Python Distribution?
5. What is PyCharm?
6. What is an identifier?
7. What is a keyword?
8. What is a variable?
9. What is an operator?
10. What is an expression?
11. What is operator precedence?
12. What is associativity?
13. What is type conversion?
14. What is the `type()` function?
15. What is the `is` operator?

- **Short Answer Questions (3 marks questions)**

1. Explain the history and features of Python.
2. Explain the advantages of Python.
3. Explain Anaconda and PyCharm as Python development environments.
4. Explain identifiers, keywords, variables and expressions.
5. Explain operators in Python.
6. Explain operator precedence and associativity.
7. Explain Python data types and type conversion.
8. Explain `type()` and `is` operators.

- **Long Answer Questions (6 marks questions)**

1. Explain the history, features and advantages of Python.
2. Explain Python programming environment, Anaconda and PyCharm.
3. Explain identifiers, keywords, statements, expressions, variables and operators with examples.
4. Explain data types, indentation, comments, input/output and type conversion in Python.
5. Explain operator precedence and associativity with examples.

Unit IV - Python Control Flow & Functions

- **Answers in one or two lines (1 mark questions)**

1. What is a control flow statement?
2. What is an **if** statement?
3. What is an **if-else** statement?
4. What is an **if-elif-else** statement?
5. What is a nested-if statement?
6. What is a **while** loop?
7. What is a **for** loop?
8. What is the **break** statement?
9. What is the **continue** statement?
10. What is a built-in function?
11. What is a user-defined function?
12. What is a function definition?
13. What is a return statement?
14. What is a default parameter?
15. What is variable scope?

- **Short Answer Questions (3 marks questions)**

1. Explain if, if-else and if-elif-else statements.
2. Explain nested-if statements.
3. Explain while and for loops with examples.
4. Explain break and continue statements.
5. Explain built-in functions and commonly used modules.
6. Explain function definition, calling and return statement.
7. Explain scope and lifetime of variables.
8. Explain default parameters.

- **Long Answer Questions (6 marks questions)**

1. Explain decision control statements in Python with suitable examples.
2. Explain while and for loops with suitable examples.
3. Explain functions in Python with function definition, calling and return statement.
4. Explain built-in functions and commonly used Python modules.
5. Explain scope and lifetime of variables with examples.